Smart Wound Dressings for Chronic Diabetic Wounds: Translating Wound-Bed Feedback into Clinician-Led Decision Support

A clinically focused medical-engineering narrative review with literature-derived evidence mapping

Running title: Smart wound dressing decision support

Article type: Structured narrative review with exploratory evidence mapping

Abstract

Chronic diabetic wounds are difficult to manage because clinically important wound-bed changes may begin before visible deterioration is obvious. Conventional dressing care remains essential, but assessment is episodic and usually requires dressing removal, visual inspection, symptoms and clinical experience. This structured narrative review asks which wound-bed signals are biologically meaningful, repeatedly measurable in dressing-compatible systems and sufficiently interpretable to support clinician-led timing decisions. Candidate literature was identified through structured searches of PubMed/MEDLINE, Scopus, Web of Science Core Collection, IEEE Xplore and Google Scholar, supplemented by backward and forward citation searching. Search blocks combined chronic or diabetic wound terms with smart-dressing or wearable-sensor terms, parameter terms including pH, temperature, moisture or exudate, impedance, oxygenation or perfusion and biomarkers, and outcome terms including continuous monitoring, early detection, wound closure and healing time. The final paper-level index contained 94 unique records; 19 studies with directly extractable longitudinal or sensor-performance evidence formed the manually verified core. Search-block yields are reported as overlapping included-study counts rather than unstable database-hit counts. Extracted time-series data were normalized to a common biological time unit and ordered in ascending time. Analyses were selected as an inference ladder: descriptive coverage to identify mature parameters; temporal trajectories to identify early and persistent change; co-occurrence, correlation and networks to select complementary sensor combinations; and PCA, t-SNE, UMAP, multidimensional scaling and clustering to examine paper-level evidence structure. These analyses support a mechanistic pathway in which earlier detection may reduce avoidable delay in clinical reassessment and thereby has potential to improve the healing trajectory. They do not by themselves prove shorter healing time. Direct comparative time-to-closure evidence and prospective clinical trials are required for that claim.

Keywords: chronic wound; diabetic foot ulcer; smart wound dressing; intelligent plaster; wound pH; wound temperature; wound moisture; clinical decision support; wound monitoring; dressing decision

1. Introduction

Chronic wounds, including diabetic foot ulcers, venous leg ulcers, pressure injuries, surgical wounds and burns, are a major clinical burden because they may heal slowly, recur, become infected, require hospitalization or progress to amputation. In diabetic wounds, neuropathy, impaired perfusion, altered immunity, pressure loading and delayed tissue repair make clinical management especially difficult [1,2].

The central clinical problem is that the wound bed can deteriorate under a dressing before obvious external signs appear. Persistent inflammation, microbial burden, hypoxia, exudate imbalance and impaired epithelialization may accumulate between two dressing changes [1,2,16]. A clinician may detect deterioration during inspection, but the biological shift may have started hours or days earlier.

Smart wound dressings aim to reduce this information gap by measuring pH, temperature, moisture/exudate, oxygenation, impedance, colour, wound area or biochemical markers at or near the wound-dressing interface [3,6-8,12,13,19-25]. These parameters are clinically useful only when converted into interpretable trends. A raw pH, temperature or impedance value without context may increase information burden rather than improve care.

This review addresses three linked questions. First, which published studies are sufficiently relevant to a clinically focused review of smart wound dressings for chronic diabetic wounds? Second, which analytical techniques can organize heterogeneous literature-derived data without implying a false meta-analysis or patient-level diagnostic model? Third, what coherent translational message emerges when wound biology, sensing technology, longitudinal evidence and clinical workflow are considered together?

Review storyline. The paper follows a clinical-to-engineering-to-clinical pathway: chronic wound instability produces biochemical, thermal and fluid changes beneath the dressing; episodic inspection can delay recognition; serial sensing makes those deviations visible; purpose-matched analyses identify a practical and complementary sensor set; and clinician-led interpretation can trigger earlier reassessment and treatment correction. The hypothesized downstream benefit is reduced avoidable delay and a better healing trajectory, with possible shortening of healing time. The review does not treat this final step as proven unless supported by direct comparative healing-outcome studies.

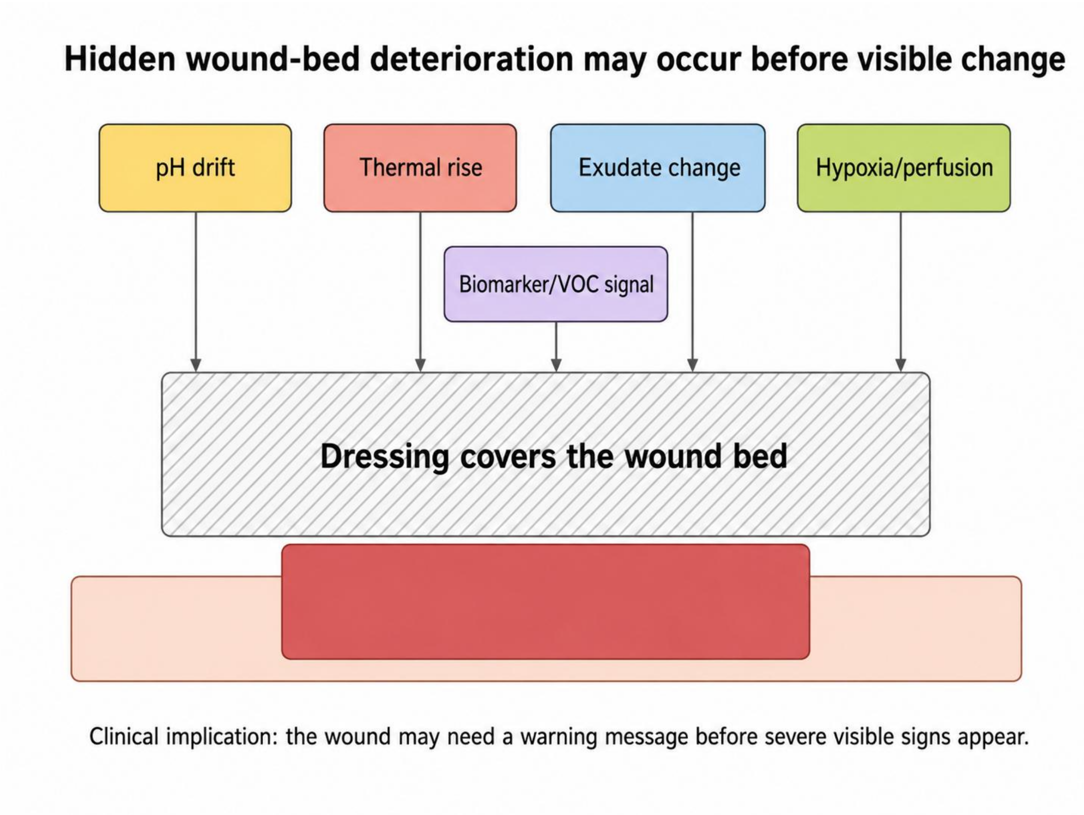

Figure 1. Hidden wound-bed deterioration under a dressing. Biochemical, thermal, moisture, oxygenation and biomarker changes may develop before visible clinical deterioration becomes obvious.

2. Clinical Background and Decision Burden

Normal wound healing is usually described through overlapping inflammatory, proliferative, epithelialization and remodelling phases. In chronic diabetic wounds, this sequence may be interrupted by infection, biofilm, ischemia, pressure, oedema, neuropathy and systemic metabolic disturbance [1,2,14-16].

The practical burden for the clinical team is decision timing. A doctor or wound-care nurse must decide whether to leave a wound undisturbed, inspect it earlier, change dressing strategy, assess infection, debride, improve offloading, refer for vascular evaluation or escalate to advanced therapy. These decisions are difficult because the wound bed is dynamic and patient-specific.

Conventional dressing care remains the foundation of wound management. Its limitation is not lack of clinical value, but intermittent information flow. Dressing removal allows direct assessment, but over-frequent removal may disturb granulation tissue, increase pain and cost, expose the wound to contamination and interrupt the moist healing environment. Delayed inspection may miss saturation, maceration, odour, infection or stalled healing.

A smart wound dressing is therefore best framed as a timing-support tool. It should not replace clinical examination, microbiological testing or vascular assessment. Its role is to provide reliable wound-bed feedback that helps clinicians decide when the wound needs attention.

Table 1. Conventional dressing care compared with smart wound dressing feedback.

Care approach	Information available	Decision value	Main limitation
Conventional dressing care	Visual inspection, exudate, odour, pain, wound size, peri-wound condition, symptoms and clinician judgement during scheduled review.	Essential for diagnosis and treatment planning; allows direct assessment and clinical reasoning.	Information is episodic and may miss hidden changes between dressing changes.
Smart wound dressing feedback	Repeated or continuous pH, temperature, moisture/exudate, impedance, oxygenation, colour,	May prompt earlier inspection, dressing reassessment or escalation when trends deviate from patient	Requires calibration, safe interpretation, workflow integration and prospective validation.

Care approach	Information available	Decision value	Main limitation
	wound-area or biomarker trends.	baseline.	

3. Current Assessment and the Need for Contextual Feedback

Current chronic wound assessment includes tissue appearance, wound size, exudate, odour, pain, peri-wound status, palpation for warmth, infection scoring, culture or laboratory testing when indicated, imaging and vascular assessment. These methods are central to wound practice and cannot be replaced by a sensor.

The main limitation is timing. A wound-size change may require days or weeks to become evident, whereas pH drift, thermal change, exudate surge or impedance shift may occur earlier. Smart monitoring can complement clinical review by translating under-dressing measurements into time trends and clinically interpretable warnings.

The safest interpretation is therefore contextual. A smart dressing should show the current trend, the parameters responsible for concern, confidence or calibration status, and the recommended clinical check. It should not display isolated numbers as diagnostic commands.

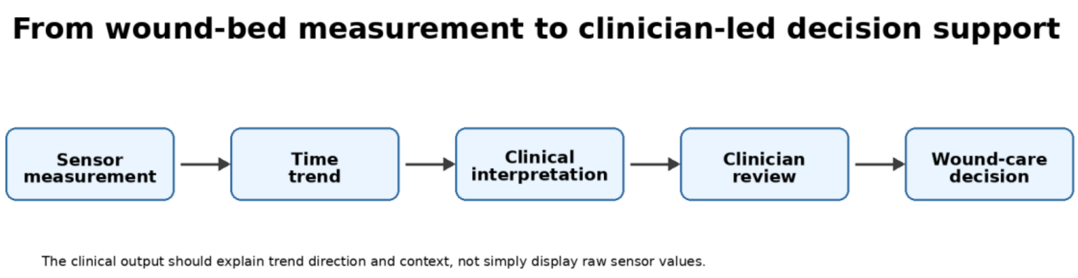

Figure 2. Sensor value to clinician-led decision-support pipeline. A clinically useful smart dressing should transform raw measurement into time trend, clinical interpretation, clinician review and wound-care decision.

4. Review Design, Search Strategy, Paper Selection and Evidence Mapping

4.1 Review design and review question

This study was designed as a structured narrative review with exploratory evidence mapping. It was not designed as a systematic review, diagnostic-accuracy review or formal meta-analysis. The review question was: among wound-bed measurements reported for chronic diabetic wounds and related wound models, which parameters are clinically meaningful, repeatedly measurable using wearable or dressing-compatible technologies, and interpretable enough to support the timing of clinician review?

The scope was defined using a population-concept-context logic. The population/problem included chronic wounds, with emphasis on diabetic foot ulcers and clinically relevant wound models. The concept included measurable wound-bed parameters and smart or wearable dressing technologies. The context was serial or continuous monitoring intended to support inspection, dressing reassessment, infection evaluation, perfusion review or other clinician-led decisions.

4.2 Information sources, search period and Boolean search strategy

Candidate studies were searched in PubMed/MEDLINE for medical and clinical literature, Scopus and Web of Science Core Collection for multidisciplinary citation coverage, IEEE Xplore for wearable electronics and sensing systems, and Google Scholar as a supplementary source for cross-disciplinary and difficult-to-index records. Backward reference checking and forward citation searching were performed from key reviews and landmark smart-dressing studies. Publisher websites were used to retrieve verified full texts and bibliographic details rather than as substitutes for the indexed database search. The search covered database inception to 16 June 2026. Search reporting was informed by PRISMA 2020 and PRISMA-S principles, while the article remained a structured narrative review rather than a claimed systematic review [36-38].

The search was built from four concept blocks. Block A described the wound condition: chronic wound*, diabetic foot ulcer*, diabetic wound*, non-healing wound*, wound infection and wound healing. Block B described the device: smart wound dressing*, intelligent wound dressing*, smart bandage*, wearable wound sensor*, flexible biosensor*, bioelectronic system* and wireless wound monitor*. Block C described measurable parameters: pH, acidity, alkalinity, temperature, thermal, thermometry, moisture, hydration, exudate, wetness, impedance, bioimpedance, oxygenation, perfusion, glucose, uric acid and biomarker*. Block D described monitoring and outcomes: continuous monitoring, real-time monitoring, early detection, clinical decision support, wound closure, wound-area reduction, healing rate and healing time. Broad searches combined A AND B; parameter-specific searches combined A AND the relevant C terms AND monitoring terms from D. This approach avoided excluding pH-only, temperature-only, moisture-only and therapeutic closed-loop studies through one overly restrictive query.

Core PubMed/MEDLINE syntax example: (("Diabetic Foot"[Mesh] OR "Foot Ulcer"[Mesh] OR "Wound Healing"[Mesh] OR chronic wound*[tiab] OR diabetic foot ulcer*[tiab] OR diabetic wound*[tiab] OR non-healing wound*[tiab]) AND (smart dressing*[tiab] OR intelligent dressing*[tiab] OR smart bandage*[tiab] OR wearable wound sensor*[tiab] OR flexible biosensor*[tiab] OR bioelectronic system*[tiab]) AND (pH[tiab] OR temperature[tiab] OR moisture[tiab] OR exudate[tiab] OR impedance[tiab] OR oxygenation[tiab] OR perfusion[tiab] OR biomarker*[tiab] OR monitor*[tiab])). Scopus searches used TITLE-ABS-KEY(...), Web of Science used TS=(...), and IEEE Xplore used All Metadata with the same concept blocks.

Table 2. Reproducible search blocks and their yield within the final manually verified core.

Query block	Database-adaptable Boolean formula	Included-study yield, n*	Contribution to the review
Q1 Broad smart/wearable wound monitoring	A AND ("smart wound dressing*" OR "intelligent wound dressing*" OR "smart bandage*" OR "wearable wound sensor*" OR "flexible biosensor*" OR "bioelectronic system*")	12	Identified integrated, flexible, wireless and dressing-compatible monitoring platforms.
Q2 Wound pH	A AND (pH OR acidity OR alkalin*) AND (sensor* OR monitor* OR profil* OR infection)	9	Captured pH sensing, pH profiling and pH-related infection or healing evidence.
Q3 Wound temperature and thermal monitoring	A AND (temperature OR thermal OR thermometr*) AND (sensor* OR monitor* OR healing OR infection OR perfusion)	8	Captured local temperature trends, thermal transport, inflammation and perfusion-related monitoring.
Q4 Moisture, hydration and exudate	A AND (moisture OR hydration OR exudate OR wetness) AND (sensor* OR dressing* OR monitor*)	2	Captured dressing saturation and wound-fluid or hydration sensing evidence.
Q5 Multimodal biomarkers and early infection detection	A AND (multimodal OR multiplex* OR biomarker* OR infection OR inflammation) AND ("early detection" OR "continuous monitoring" OR "real-time monitoring")	5	Captured studies combining two or more signals for early-warning or infection-related interpretation.
Q6 Closed-loop treatment and healing outcomes	A AND ("closed loop" OR theranostic OR "drug delivery" OR "electrical stimulation" OR heating OR "negative pressure") AND ("wound closure" OR "area reduction" OR "healing rate" OR "healing time")	8	Captured systems linking monitoring or active treatment with wound-healing outcomes.

***Note:** A denotes the wound-condition block defined above. Included-study yields are non-additive because one paper could satisfy several query blocks. These values map the final 19-paper manually verified core to the search strategy; they are not presented as total database-result counts.

4.3 Eligibility and paper-selection criteria

Studies were eligible when they addressed chronic wounds, diabetic foot ulcers or a clinically relevant wound model and contributed to at least one of four evidence domains: (1) wound biology or clinical decision burden; (2) measurement of a chemical, thermal, moisture, electrical, optical or biochemical wound parameter; (3) longitudinal healing, infection, wound-area or intervention outcomes; or (4) engineering-to-clinical translation, including wearable integration, calibration, safety, workflow or clinician interpretation. Human, animal, ex vivo and in vitro studies were eligible, but study design was recorded and interpreted separately. Primary research was prioritized for quantitative extraction; focused reviews were used for context, terminology and citation chasing rather than treated as independent clinical observations.

Duplicate records, non-wound sensing studies, editorials, conference abstracts without sufficient methods or results, commercial product pages, unverifiable reports, and papers without a measurable wound parameter, clinically interpretable outcome or relevant translational contribution were excluded. Therapeutic dressing studies were retained only when they reported wound closure, healing rate, wound-area trajectory, infection-related change, temperature, pH or another wound-microenvironment outcome relevant to the review question. English-language full text was required for manual extraction.

4.4 Screening, deduplication and selection for synthesis

Records were deduplicated using DOI, exact and near-exact title, first author and publication year. Screening then proceeded in two stages: title and abstract assessment against the wound, device, parameter and outcome concepts, followed by full-text eligibility assessment. Exclusion reasons were recorded as duplicate, not wound-related, no relevant sensing or outcome data, insufficient full text, or no direct contribution to the clinical-to-engineering review question. The final paper-level source index contained 94 unique papers for evidence mapping. Nineteen papers containing directly extractable longitudinal, sensor-performance or healing-outcome data formed the manually verified core used for high-confidence extraction.

Three evidence sets were kept distinct: the 19-paper manually verified extraction core, the 94-paper paper-level evidence index used for coverage and latent-structure mapping, and the broader set of sources cited for clinical background, methods and interpretation. These sets overlap but are not interchangeable. The 94-paper index is therefore not a patient sample size, and the 19-paper core is not presented as the complete universe of smart wound-dressing research.

4.5 Data extraction, provenance and confidence control

The extraction database recorded source identity, study or model type, wound context, parameter domain, time point, measurement value and unit where available, extraction route, confidence level and interpretive notes. Manual extraction was used for the 19-paper core. Automatic text extraction and graph-derived candidate values expanded coverage in the wider index, but their provenance was retained so that automatically obtained values could not be mistaken for manually verified data.

The full extraction set contained 12,799 candidate evidence rows: 1,106 rows (8.6%) from the manually curated 19-paper subset, 11,689 rows (91.3%) from automatic text extraction, and 4 coverage placeholders for non-extractable studies. Consequently, 75 of the 94 papers had no manually checked rows in the current database. The time-series layer contained 12,209 usable time-value candidate rows from 76 papers, and a stricter chart-ready subset contained 4,352 numeric time-value rows.

Time was converted to a common biological time unit where the source allowed conversion and values were sorted in ascending time before trajectory analysis. Missing values were not imputed. The candidate corpus contained 912 high-confidence rows (7.1%), 7,953 medium-confidence rows (62.1%), 3,930 low-confidence rows (30.7%) and 4 rows marked not usable or non-extractable. Counts therefore reflect reporting intensity, extraction route and extractability, not clinical effect size.

4.6 Selection of analytical techniques

Analytical techniques were selected as a staged inference ladder aligned with the paper's clinical story, rather than to maximize computational complexity. Level 1, descriptive frequency and missingness, identified which parameters were sufficiently represented to support a practical sensor set. Level 2, time-ascending trajectories and curve features, examined whether measurable changes appeared early, persisted, recovered or preceded slower wound-area outcomes. Level 3, Jaccard co-occurrence, Spearman association and network mapping, identified complementary parameter combinations for multimodal monitoring and helped distinguish useful sensor fusion from redundant measurement. Level 4, PCA, t-SNE, UMAP, multidimensional scaling and hierarchical clustering, examined whether similar paper-level evidence profiles formed reproducible linear or nonlinear neighbourhoods. Level 5 is the clinical outcome layer: direct comparison of time to complete healing, wound-area reduction or treatment escalation between intelligent-system and standard-care groups.

The first four levels can support parameter prioritization, early-warning plausibility, sensor fusion and future trial design, but they cannot prove that an intelligent dressing shortens healing time. A defensible healing-time claim requires comparable intervention and control studies. Time-to-event outcomes should be synthesized using hazard ratios or other survival-analysis estimates; continuous healing duration may be summarized using mean differences or standardized mean differences; and complete-closure proportions may be summarized using risk ratios. Pooling should be restricted by wound type, study design, intervention and follow-up. Because the present database mixes human, animal, ex vivo and in vitro evidence, healing-time reduction is treated as a downstream hypothesis: earlier detection may reduce avoidable clinical delay and has potential to improve healing trajectories, but prospective comparative trials are required to demonstrate the endpoint.

4.7 Narrative synthesis and the story of the review

The narrative synthesis was organized around a causal translation pathway rather than a catalogue of devices. Chronic diabetic wounds can remain inflamed, infected, ischemic, over-hydrated or mechanically stressed between scheduled inspections. Delayed recognition can delay debridement, antimicrobial review, offloading correction, vascular assessment or dressing adjustment, each of which may prolong the healing course. Smart sensing is positioned as an information intervention: pH, temperature, moisture, impedance and biomarker trends may reveal deterioration earlier; the analytical ladder identifies which signals are feasible, temporally informative and complementary; and clinician-led interpretation converts those trends into earlier reassessment. The central conclusion is therefore carefully bounded: intelligent monitoring may reduce avoidable decision delay and may thereby improve healing trajectories, but the present exploratory evidence mapping does not independently establish a reduction in healing time.

Box 1. Transparent paper-selection framework used in this review.

Selection domain	Included	Excluded or restricted
Clinical scope	Chronic wounds, diabetic foot ulcers and clinically relevant wound models.	Non-wound sensing studies or conditions without a wound-monitoring link.
Evidence contribution	At least one measurable wound parameter, longitudinal wound outcome, sensor method or decision-support issue.	No wound-relevant parameter, outcome or translational contribution.
Study design	Human, animal, ex vivo, in vitro and focused reviews; design recorded and interpreted separately.	Mixed designs treated as equivalent patient evidence.
Therapeutic studies	Retained only when they supplied usable wound trajectory or microenvironment data.	Pure treatment descriptions without relevant monitoring or longitudinal data.
Source quality and duplication	Full-text research evidence with verifiable title/authorship/year/DOI where available.	Duplicate copies, product pages, inaccessible/insufficient records and unverifiable items.

Table 3. Literature-derived database layers and interpretation limits.

Evidence layer	Count	Correct interpretation	Incorrect interpretation
Paper-level index	94 unique papers	Source index and parameter-presence matrix.	Not patient sample size or clinical cohort size.
Full candidate evidence set	12,799 rows	Candidate extracted text/table evidence from manual and automatic sources.	Not all rows are clinically validated numerical observations.
Manual versus automatic provenance	1,106 manual rows from 19 papers; 11,689 automatic rows; 4 placeholders	Shows manually checked core plus larger automatic candidate corpus.	Automatic rows are not a manually verified meta-analysis.
Confidence summary	912 high; 7,953 medium; 3,930 low; 4 not usable	Quality distribution for candidate evidence rows.	Medium/low rows are insufficient for threshold derivation without manual checking.
Time-series layer	12,209 time-value candidate rows from 76 papers	Normalized biological time and measurement value for broad evidence mapping.	Not identical to the full 94-paper source index.
Chart-ready subset	4,352 strict numeric XY rows	Tighter numeric subset for plotted time-vs-parameter figures.	Not the authoritative total of all usable time-value evidence.
Structure-mapping profiles	94 paper-level profiles	Evidence-profile matrix for PCA/t-SNE/UMAP/MDS and clustering.	Not patient phenotypes, diagnostic classes or wound subtypes.

5. Measurable Wound-Bed Parameters and Clinical Meaning

Measurable wound-bed parameters should be interpreted by clinical decision value, not technology novelty alone. Each parameter has a biological meaning, a possible decision role and a measurement limitation. No single parameter is sufficient for wound-state interpretation; combined trends are safer and more clinically useful.

5.1 Wound pH

Wound pH reflects the chemical wound-bed environment, including acid-base state, tissue repair conditions and microbial context. Healing wounds are often discussed as moving toward a more favourable acidic-to-neutral environment, whereas chronic or infected wounds may show persistent neutral or alkaline behaviour [3-5,19,22,30,31]. pH can support reassessment of wound-bed preparation, dressing choice and antimicrobial strategy, but pH alone should not diagnose infection. Sensor limitations include calibration drift, biofouling, wound-fluid heterogeneity and dressing interaction.

5.2 Wound temperature

Temperature is practical for continuous monitoring because local thermal change may reflect inflammation, infection risk, perfusion recovery or impaired tissue activity [6-10,26]. Relative change from patient baseline, adjacent skin or contralateral limb is safer than a universal cut-off. A rise may indicate inflammation in one context but perfusion recovery in another; ischemic or neuropathic diabetic feet may also show a blunted thermal response. Absence of a thermal signal must therefore never exclude infection or ischemia.

5.3 Moisture and exudate

Moisture and exudate influence epithelialization, dressing performance, maceration and bacterial burden. Controlled moisture supports healing, excessive wetness may saturate dressings and damage peri-wound tissue, and excessive dryness may delay epithelial migration. Moisture can be monitored through impedance, capacitance, resistance, hydration sensors, optical signals or wetness indicators [7,12,13]. Thresholds are dressing-specific and wound-type-specific.

5.4 Oxygenation and perfusion

Oxygenation and perfusion support collagen synthesis, angiogenesis, immune function and infection resistance. Diabetic or ischemic wounds may fail to heal when oxygen delivery is poor. Oxygen/perfusion monitoring may support vascular assessment when combined with low temperature, delayed wound-area reduction or clinical ischemic signs [10,21]. Calibration and correlation with vascular assessment are essential.

5.5 Wound area and healing percentage

Wound area, closure percentage and healing rate are slower than biochemical signals but remain core outcome anchors. A plateau in wound reduction may support reassessment of infection, pressure, vascular supply, dressing type or debridement need. Digital imaging can improve quantification, but lighting, scale, camera distance and segmentation accuracy must be controlled.

5.6 Electrical impedance and related outputs

Impedance, resistance, capacitance, conductivity, voltage or current outputs can reflect tissue hydration, ionic wound fluid, electrode contact and dressing saturation [12,13]. These outputs are useful engineering proxies, but clinical interpretation depends on electrode geometry, frequency, wound-fluid conductivity, dressing material and contact stability.

5.7 Colour, optical features and biomarkers

Colour, imaging, fluorescence and biochemical markers can provide information about granulation, slough, necrosis, erythema, epithelialization, inflammation, metabolism or microbial activity [14,20,21,25]. These signals are promising but require standardization, selectivity, calibration and validation against clinical outcomes before use as decision-support inputs.

Table 4. Wound parameters and clinician-facing interpretation.

Parameter	Clinical meaning	Helpful trend	Warning trend	Key limitation
pH	Chemical wound-bed environment and microbial/tissue-repair context.	Stable or moving toward favourable wound-bed range.	Persistent abnormal drift, especially alkaline shift with other warning signs.	Drift, fouling, wound-fluid variability; not diagnostic alone.
Temperature	Local inflammation, perfusion and thermal activity.	Stable patient-specific baseline or improving recovery pattern.	Sustained rise, abnormal gradient or poor recovery.	Ambient, contact, insulation and anatomical-site effects.
Moisture/exudate	Hydration, dressing saturation and maceration/dryness risk.	Controlled moist environment.	Rapid wetness increase, saturation or excessive dryness.	Thresholds depend on dressing and sensor design.
Oxygenation/perfusion	Tissue oxygen supply and vascular support.	Stable or improving oxygenation/perfusion.	Low oxygenation with delayed closure or low temperature.	Needs correlation with vascular assessment.
Wound area/healing	Macroscopic healing outcome.	Area reduction and closure progression.	Plateau, enlargement or delayed closure.	Changes slowly; image standardization required.
Electrical impedance	Hydration, tissue-fluid and dressing interface.	Expected hydration trend for dressing type.	Sudden impedance/resistance/capacitance change.	Device, frequency and electrode-geometry dependence.
Optical/biomarker features	Tissue appearance, inflammation, metabolism and microbial activity.	Improving granulation/epithelialization or decreasing inflammatory signals.	Necrosis, slough, abnormal erythema, high ROS/VOCs/cytokines or microbial signal.	Lighting, selectivity, fouling and validation burden.

6. Analysis Strategy, Evidence Mapping and Time-Ascending Parameter Analysis

The analysis plan was selected to answer five linked questions: which wound parameters dominate the available evidence; how reported measurements change with biological time; which parameters repeatedly appear together and may be complementary; whether papers form reproducible evidence neighbourhoods; and what additional outcome evidence would be required to demonstrate shorter healing time. The first four questions support the proposed early-warning mechanism and system architecture. The fifth requires direct controlled outcome evidence and is not answered by latent-space visualization alone.

Descriptive frequency and missingness analysis addressed evidence maturity. Sensor performance or calibration, bacterial or infection evidence, electrical or bioimpedance outputs, temperature, wound area or closure, and pH were among the strongest reporting groups in the 94-paper corpus. These counts are extraction-weighted and confidence-dependent; they support prioritization for future validation but do not establish clinical reliability, diagnostic accuracy or treatment effectiveness.

Time-ascending plots and trajectory features addressed temporal interpretation. Normalized biological time was placed on the X-axis and measurement value on the Y-axis, reducing extraction-order bias and allowing baseline, final value, slope, range, variability and area-under-curve summaries. The central clinical implication is that an isolated pH, temperature or moisture value is less useful than the direction, magnitude, variability and persistence of change.

Jaccard co-occurrence, Spearman correlation and parameter-network analysis addressed complementarity. Jaccard similarity was appropriate for binary paper-level parameter presence, whereas Spearman correlation was used for monotonic associations without assuming normal distributions in heterogeneous evidence counts. Network mapping made recurrent combinations visible. These patterns suggest sensor-fusion candidates but represent evidence structure rather than biological causality.

PCA, t-SNE, UMAP, multidimensional scaling and hierarchical clustering addressed global and local paper-level structure. PCA supplied interpretable linear variance and feature loadings; the nonlinear methods tested neighbourhood stability under different assumptions; and clustering supplied a tree-based comparison. The outputs are research-neighbourhood maps, not patient phenotypes, diagnostic classes or treatment-response groups. A 7-day measurement in an animal model, a human diabetic foot ulcer cohort and an in vitro calibration experiment do not represent the same biological context and were therefore not interpreted as interchangeable.

Table 5. How each analytical method supports the review storyline and its claim boundary.

Method	Question answered	Contribution to the healing-time pathway	Claim boundary
Parameter frequency and missingness	Which parameters are sufficiently represented and extractable?	Selects a feasible minimum sensor set and prevents the proposed system from depending on poorly supported signals.	Shows evidence maturity, not faster healing.
Time-ascending trajectories and curve features	Which signals change early, persist, recover or precede slower closure outcomes?	Supports the early-warning mechanism by locating measurable deviations before visible or area-based change.	Does not show that acting on the warning improves outcome.
Jaccard, Spearman and network mapping	Which parameters are co-reported, complementary or redundant?	Supports multimodal sensor fusion that may improve warning specificity and reduce missed deterioration or unnecessary alerts.	Association and co-reporting are not biological causation.
Principal component analysis	Which variables explain the dominant linear variation in paper-level evidence profiles?	Identifies a smaller interpretable feature set for device design and future clinical risk models.	Does not create a validated patient score or treatment effect.
t-SNE, UMAP, MDS and hierarchical clustering	Do similar evidence profiles form stable local or global neighbourhoods under different assumptions?	Generates hypotheses about study or wound-state stratification for future prospective validation.	Clusters are not validated wound phenotypes, diagnoses or response classes.
Direct comparative healing-outcome synthesis	Do intelligent or closed-loop systems shorten time to closure or improve area-reduction rate versus standard care?	This is the required endpoint layer for a genuine healing-time claim, using comparable controlled studies and time-to-event or effect-size analysis.	Not established by the present heterogeneous exploratory mapping; prospective trials or a restricted meta-analysis are required.

Taken together, the analyses support the proposed pathway at the level of plausibility and design: recurrent parameters identify what can be measured; temporal analysis identifies what may change early; multimodal mapping identifies which signals should be combined; and latent-space analysis identifies where future stratification may be useful. The resulting system could reduce avoidable delay only when an interpretable warning leads to effective clinician action. Healing-time reduction remains a clinical outcome to be tested, not an output inferred from PCA, UMAP or correlation plots.

7. Infection, Inflammation and Chemical Wound-Bed Imbalance

Infection and inflammation should not be inferred from one sensor. Bacterial or fungal activity may alter pH, temperature, exudate, odour, colour, impedance and biomarkers, but these signals can also change with non-infectious inflammation, dressing effects, perfusion recovery or normal healing.

A safer clinical strategy is to interpret infection risk as converging evidence. Alkaline pH drift, sustained local temperature change, increasing exudate, impedance shift, optical deterioration and delayed wound-size reduction may justify earlier clinical review when they occur together. This still does not diagnose infection; it prioritizes the wound for inspection and confirmation.

pH-guided interpretation should be written as support for clinical reassessment, not an automatic instruction to apply neutralizing agents or medicines. The measurement should strengthen clinical reasoning while preserving clinician responsibility.

8. Engineering-to-Clinical Translation

Clinical translation requires engineering reliability and interpretability. Sensors must remain accurate despite exudate, biofouling, motion, pressure, dressing compression, temperature variation and long wear time. The wound-contact layer must be biocompatible and comfortable, and the electronics must be isolated from direct wound fluid where possible.

pH, electrochemical and biochemical sensors are vulnerable to calibration drift and fouling. Moisture and impedance thresholds depend on dressing material, electrode geometry and wound-fluid conductivity. Temperature sensors require stable contact and ambient compensation. Optical systems require controlled lighting and scale. Biomarker sensors require selectivity and stability.

The interface should be designed for clinicians. Instead of many disconnected values, it should show the current trend, the parameter responsible for concern, calibration or confidence status, and the recommended clinical check. Patient-specific baselines are likely essential because wounds differ by anatomy, perfusion, diabetes status, dressing type, wound age and treatment history.

8.1 Tropical and Resource-Limited Deployment

Deployment in Sri Lanka and similar South Asian settings requires attention to high ambient temperature, high humidity, perspiration, dust, wet dressing environments and variable clinic conditions. These factors can confound temperature, impedance, capacitance, adhesive contact, optical interpretation and electrochemical stability.

A realistic first-stage device may not be a fully integrated disposable wireless array. A practical pathway may begin with low-cost colourimetric or hydrogel pH indicators, simple temperature sensing, absorbency or saturation indicators, smartphone-assisted imaging and a reusable reader module. More complex thermal arrays, electrochemical multiplexing and closed-loop systems may be reserved for specialist clinics or research trials until cost, durability, sterilization and reimbursement are resolved.

8.2 Claim Control for Clinical Deployment

The central claim should remain conservative and mechanistically explicit: smart wound monitoring may detect clinically relevant deviation earlier, support earlier clinician-led reassessment and reduce avoidable decision delay. Shorter healing time is a possible downstream consequence only when the warning leads to an effective change in care. This review does not prove that monitoring alone shortens healing time. Healing duration also depends on perfusion, infection burden, glycaemic control, pressure and offloading, nutrition, comorbidities, adherence and quality of clinical care. Thresholds and alert pathways must therefore be validated against clinician-labelled outcomes and comparative healing endpoints.

9. Authors' Proposed Smart Wound Dressing Concept

The proposed translational step is to develop and validate a smart wound dressing/intelligent plaster for chronic diabetic wounds. The initial prototype should focus on pH, temperature and moisture/exudate because these parameters are biologically meaningful, repeatedly measurable at the wound-dressing interface and directly linked to wound-bed chemistry, inflammation/perfusion and dressing performance.

The device should use a low-cost flexible sensor patch with a safe, non-irritant wound-contact interface. It should avoid pressure points, especially in diabetic feet. The sensing region should remain sufficiently coupled to wound fluid or the wound-dressing interface to capture useful trends, while electronic components should be protected from fluid contact. A reusable external reader or smartphone-linked module may be more realistic than fully disposable electronics in resource-limited settings.

Validation should begin with patient-specific baselines rather than universal thresholds. pH, temperature and moisture trends should be compared with clinician-labelled outcomes such as healing progression, dressing saturation, maceration, suspected infection, debridement need, offloading change and vascular review. Prospective studies are required to determine whether the device improves timing of clinical review without increasing unnecessary dressing disturbance or alarm fatigue.

10. Research Gaps and Future Directions

The main gap is no longer only sensor fabrication. Many studies show that pH, temperature, moisture, impedance, oxygenation, biomarkers and optical features can be measured. The larger translational gap is whether these measurements can be validated as safe, actionable and clinician-friendly information in real wound-care pathways.

Future studies should report raw time-stamped wound-parameter data, wound type, dressing type, treatment events, infection status, debridement, offloading, perfusion assessment and clinical outcomes. This will allow models to distinguish infection, inflammation, ischemia, maceration, dryness and stalled healing using clinician-labelled trajectories.

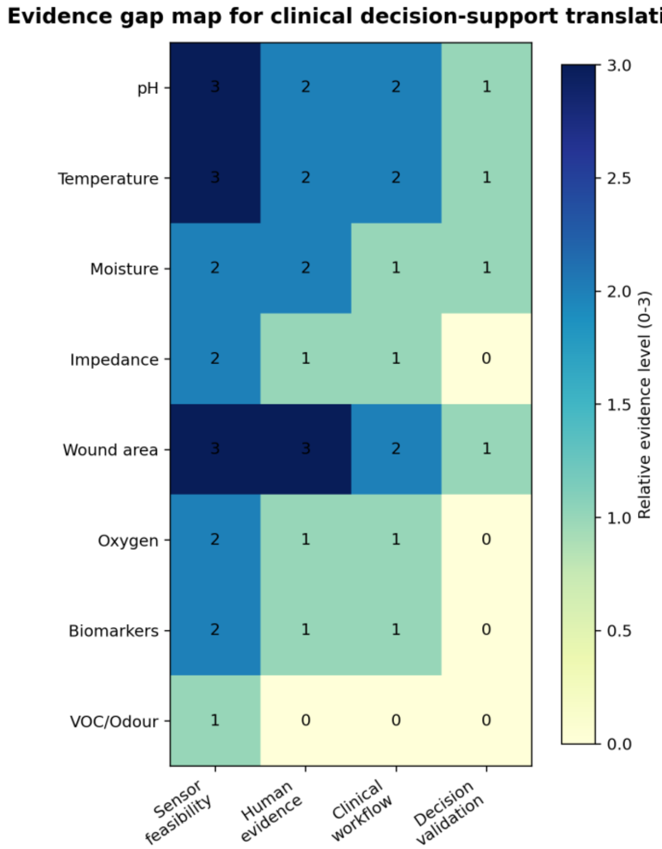

Figure 3. Evidence gap map for clinical decision-support translation. Many parameters have proof-of-concept sensor evidence, but workflow validation and human clinical decision-support testing remain limited.

Table 6. Key research gaps and future directions.

Current gap	Why it matters clinically	Validation need	Future direction
Lack of validated thresholds	Doctors need safe action limits.	Prospective threshold validation against clinical outcomes.	Develop patient-specific alarm ranges.
Single-parameter focus	Wounds are multi-factorial.	Validate multi-parameter patterns.	Combine pH, temperature, moisture, impedance and wound-area trends.
Poor time-stamped reporting	Trend interpretation needs a reliable time axis.	Standardized supplementary datasets.	Publish raw longitudinal parameter data.
Limited clinical trials	Sensor proof does not equal clinical utility.	Human wound-care pathway studies.	Test alarms in real clinics.
Notification fatigue	Too many warnings reduce usability.	Clinician usability testing.	Use tiered alerts and confidence scoring.
Weak workflow integration	Clinicians need actionable messages.	Dashboard and documentation validation.	Integrate with wound-care records.
Overlapping causes	Infection, inflammation, ischemia and maceration can appear similar.	Outcome-labelled datasets.	Train explainable clinician-in-the-loop models.
Regulatory and safety gaps	Warnings influence clinical action.	Risk management, cybersecurity and governance review.	Develop clinician-in-the-loop regulatory strategy.

11. Limitations and Data-Use Statement

This is a structured narrative review with exploratory evidence mapping rather than a prospectively registered systematic review or formal meta-analysis. The search concepts, Boolean formulas, eligibility criteria and staged screening process are reported to improve reproducibility, but the review combines clinical, animal, ex vivo, in vitro and engineering evidence for different purposes. The included-study yields in Table 2 are overlapping mappings of the manually verified core and should not be interpreted as database prevalence or effect-size evidence. A future outcome-focused systematic review should restrict eligibility to comparable controlled studies and preserve database-specific exports, screening decisions and risk-of-bias assessments.

This review is limited by study-design heterogeneity. Human clinical studies, animal experiments, ex vivo work and in vitro sensor tests may share similar time values but do not represent the same biological condition. Future meta-analyses should stratify evidence by study type before deriving clinical thresholds or recommendations.

The quantitative database contains candidate values from published studies. Some values came from automatic extraction or graph digitization and require manual verification before formal meta-analysis. The evidence maps are suitable for hypothesis generation and sensor-priority discussion, not patient-level diagnostic validation.

Latent-space methods such as PCA, t-SNE, UMAP and MDS show hidden structure in paper-level evidence profiles, but they do not prove wound phenotypes, infection classes or treatment-response categories. Clinical thresholds remain context-dependent and affected by wound type, anatomical site, perfusion, dressing material, sensor drift, biofouling, comorbidities and treatment history.

12. Conclusion

The central story of this review is a direct translation pathway. Hidden wound-bed change creates an information gap between scheduled inspections; serial sensing can make that change visible; a reproducible search and purpose-matched analytical ladder identify a practical multimodal core; and clinician-led interpretation can convert patient-specific trends into earlier reassessment. The reviewed evidence supports pH, temperature and moisture or exudate as an initial core, supported where possible by wound-area or healing trajectory, impedance, oxygenation and infection-related markers. No single parameter defines wound status.

pH reflects wound-bed chemistry and microbial/tissue-repair context. Temperature reflects local inflammation and perfusion but must be interpreted relative to patient baseline and anatomical site. Moisture/exudate is clinically important because it maps directly to dressing saturation, maceration risk and wound-bed hydration. Together, these parameters can support earlier clinician-led reassessment when trends deviate.

The literature-derived frequency, time-ascending, co-occurrence, correlation, network and latent-space analyses support the feasibility and logic of an early-warning system. They identify recurrent parameters, temporally informative trends and complementary combinations that may reduce avoidable delay in clinician review. They do not prove that healing time is shortened. That conclusion requires direct comparison of time to closure, area-reduction rate or complete-healing probability under intelligent-system and standard-care pathways.

The proposed translational output is a clinician-in-the-loop early-warning framework: stable trends may support continued care, early deviation may prompt watchful reassessment, sustained multi-parameter change may prompt inspection or dressing reassessment, and high-risk deterioration may prompt urgent clinical review. These levels are intended to prioritize attention, not diagnose, prescribe or act automatically.

Prospective, patient-specific, multi-parameter studies are required before thresholds or alarms can be used clinically. In diabetic feet, especially ischemic or neuropathic wounds, the absence of a warning must never be used to exclude infection or ischemia. The value of smart wound monitoring is to translate hidden wound-bed changes into timely, trustworthy information that supports clinical judgement.

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Supplementary Appendix A. Selected Time-Ascending Evidence-Mapping Figures

The following figures are retained as supplementary evidence maps. They should be interpreted as exploratory literature-structure visualizations rather than clinical diagnostic outputs.

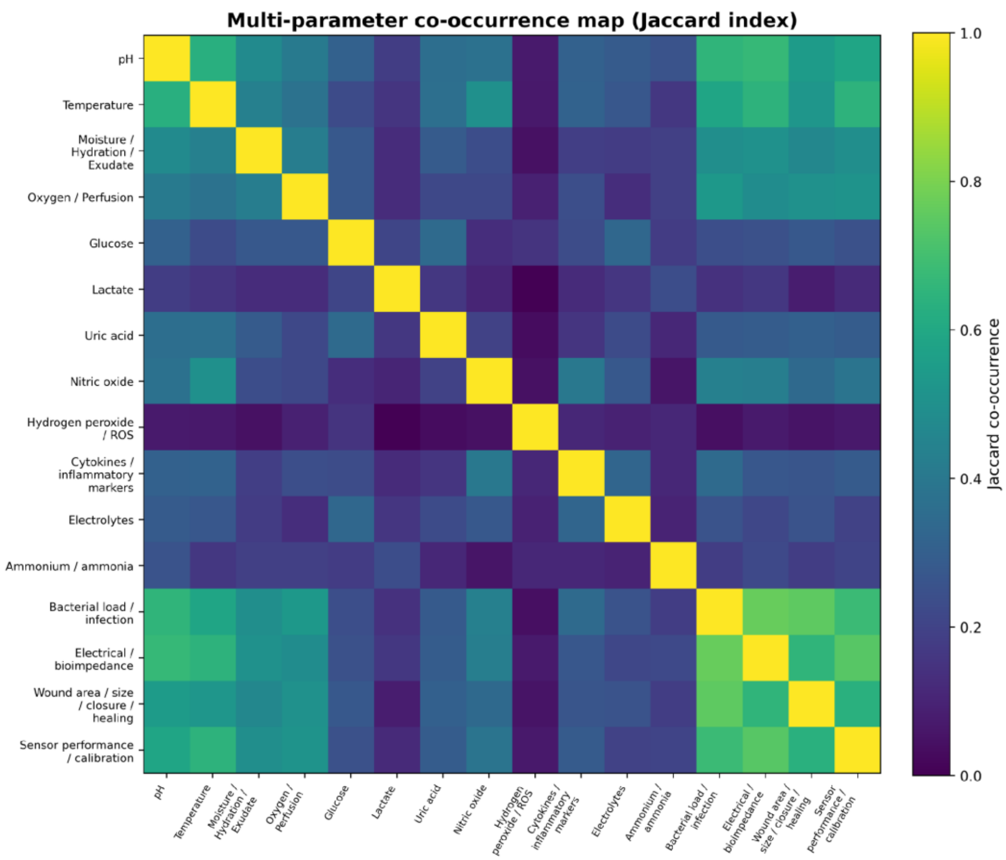

Figure S1. Multi-parameter co-occurrence heatmap based on Jaccard similarity. Higher values indicate stronger co-reporting across reviewed literature sources.

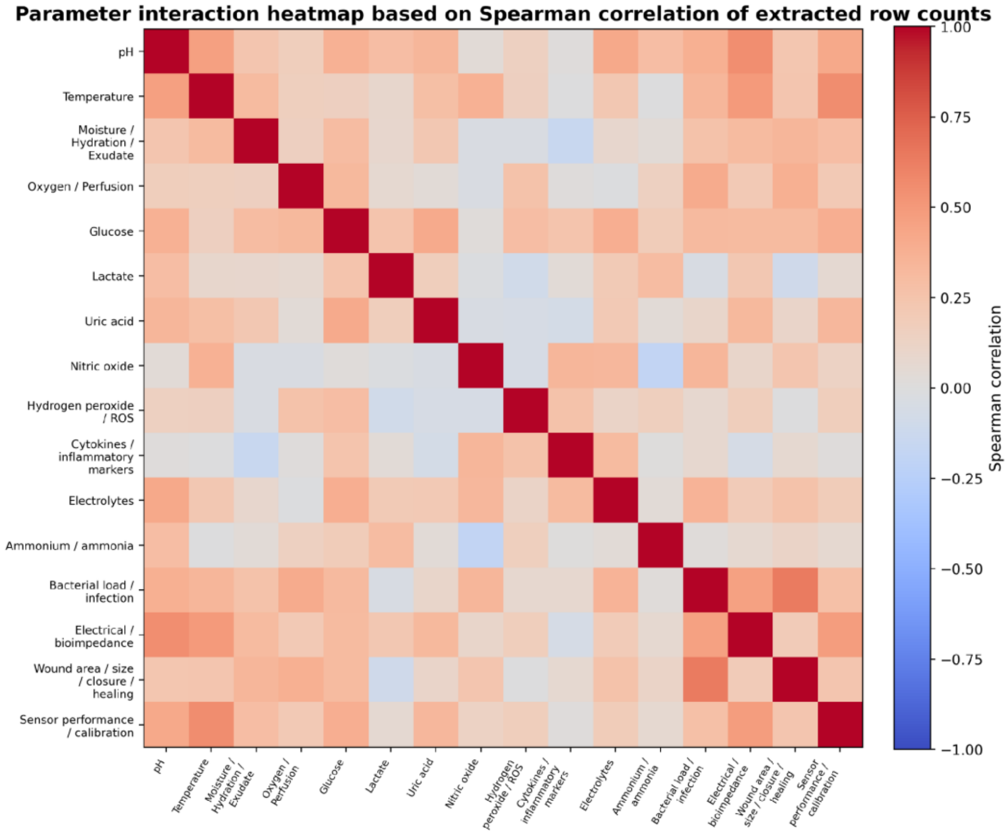

Figure S2. Parameter interaction heatmap based on Spearman correlation of extracted evidence-level row counts. The figure shows evidence-structure associations, not biological causality.

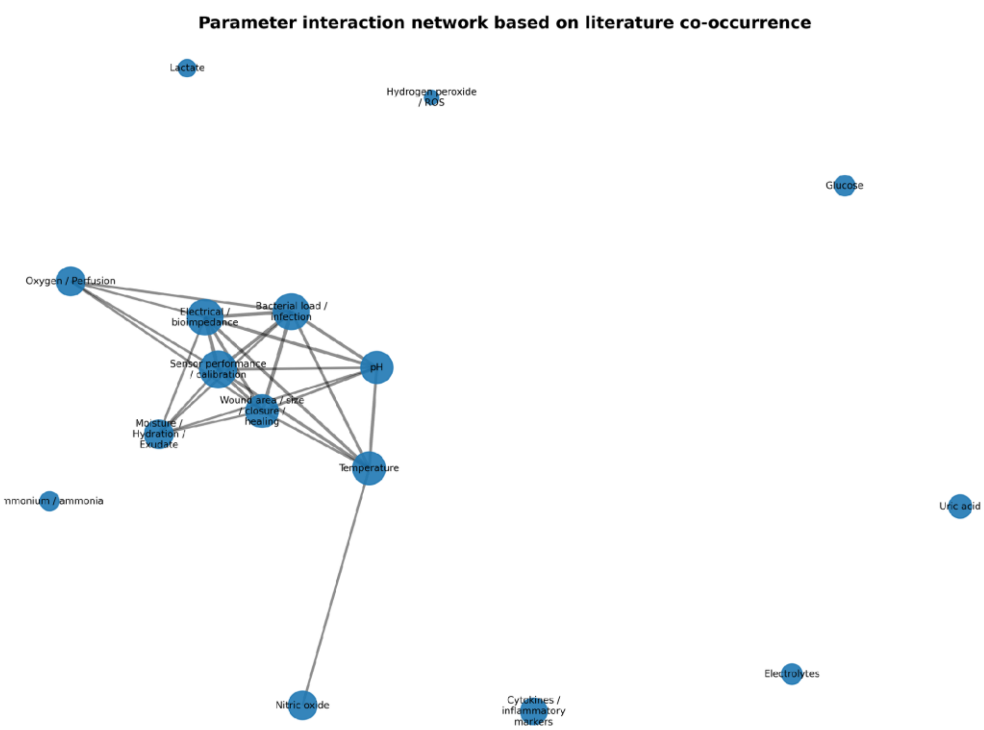

Figure S3. Parameter interaction network based on literature co-occurrence. Node position and connectivity reflect reporting and extraction patterns rather than proven clinical superiority.

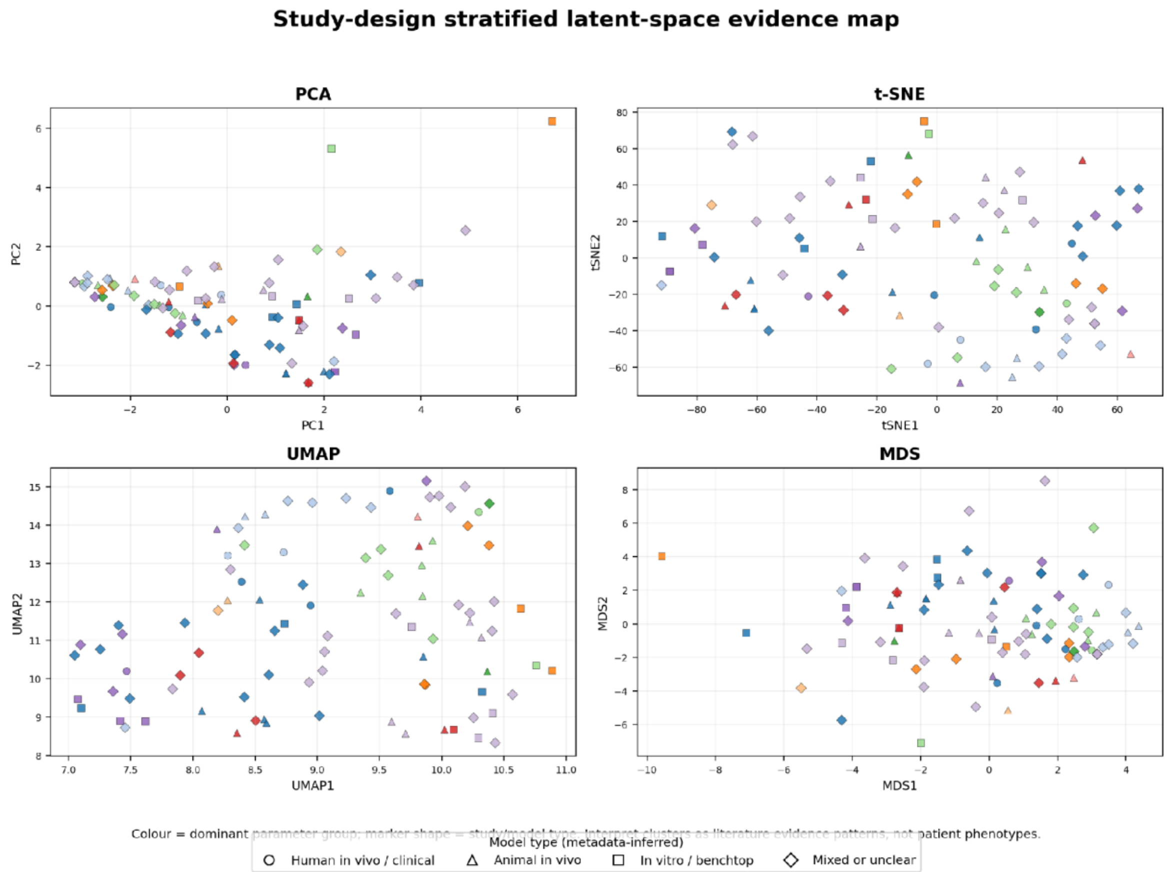

Figure S4. Study-design-stratified latent-space evidence map using PCA, t-SNE, UMAP and MDS. The plots are literature evidence maps, not patient phenotype clusters.

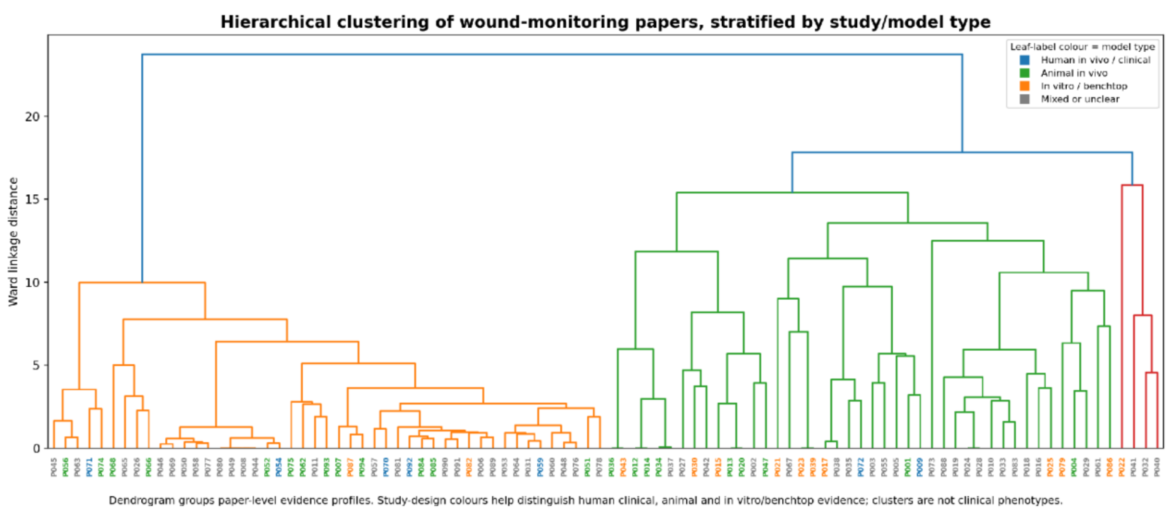

Figure S5. Hierarchical clustering of wound-monitoring papers stratified by study/model type. The branching structure shows paper-level evidence-profile similarity, not clinical wound subtypes.

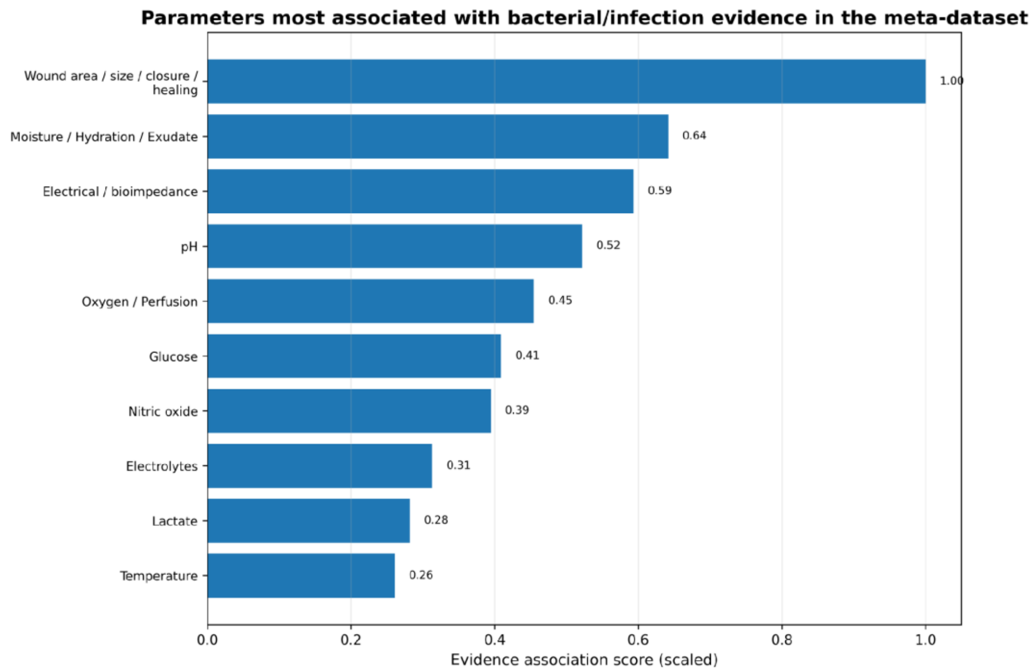

Figure S6. Infection- and inflammation-related parameter ranking in the evidence map. Ranking is influenced by extraction frequency and confidence level and should be treated as hypothesis-generating.

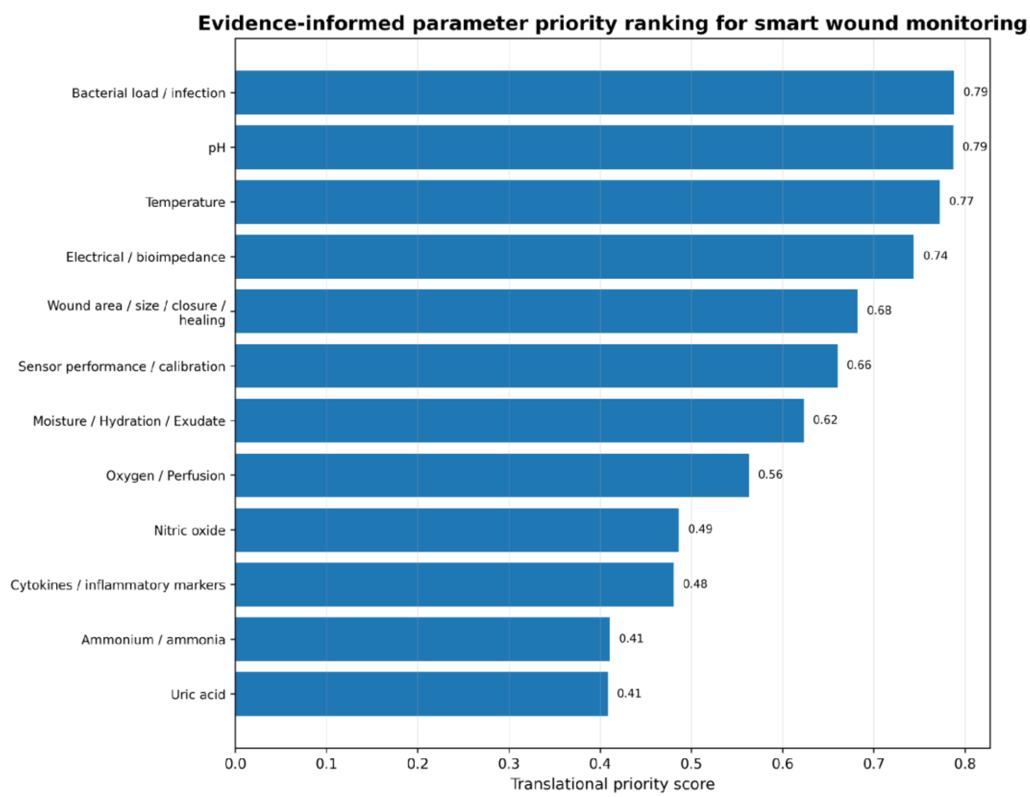

Figure S7. Evidence-informed parameter priority ranking for smart wound monitoring. Priority reflects practical relevance, interpretability, extraction confidence and likely decision value.

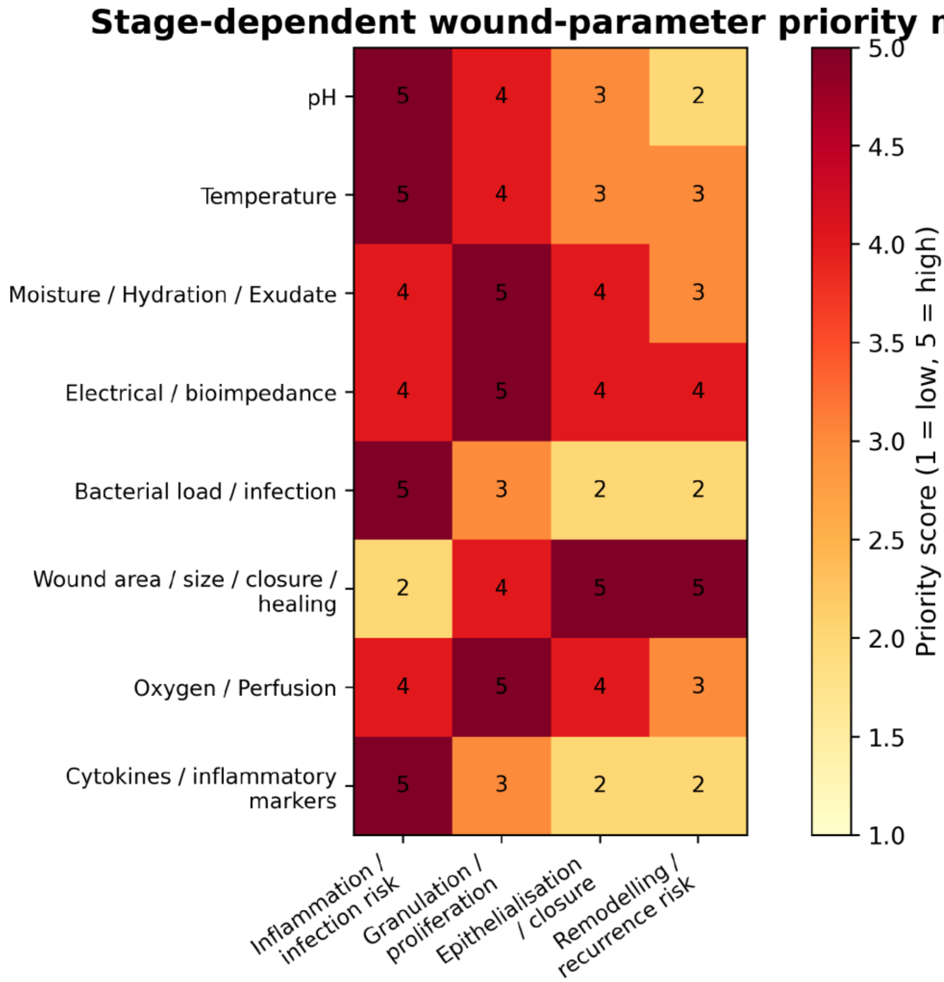

Figure S8. Stage-dependent wound-parameter priority map linking monitoring domains to inflammation/infection risk, granulation/proliferation, epithelialization/closure and remodelling/recurrence risk.